Starter: TCP/IP Prework

1. What is a protocol?

2. What does TCP/IP stand for?

3. How many layers are there in the TCP/IP protocol stack? Can you name them?

Starter: TCP/IP Prework

1. What is a protocol?

Pre-agreed signals, codes and rules to be used for data and information exchange between systems

2. What does TCP/IP stand for?

Transmission Control Protocol / Internet Protocol

3. How many layers are there in the TCP/IP protocol stack? Can you name them?

Four: Application; Transport; Network; Link;

TCP/IP

Keywords: TCP; IP; Application; Transport; Link; Network; MAC Address; Port;

TCP/IP

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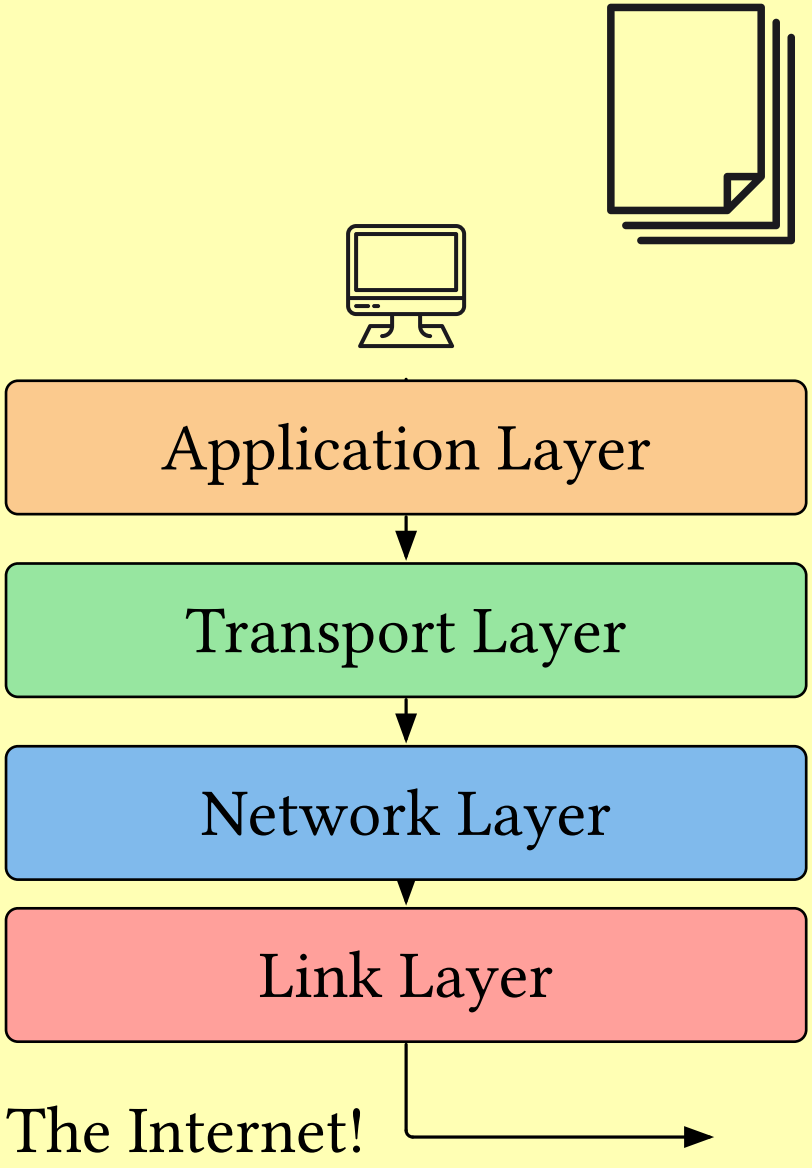
The TCP/IP Stack

In order for a device on the internet (**host**) to transfer a file from a program, , to a program on another device, we use this! The TCPIP stack.

Data from the program is passed into the Application layer, which does "stuff", then passes the result onto the Transport Layer. (and so on...)

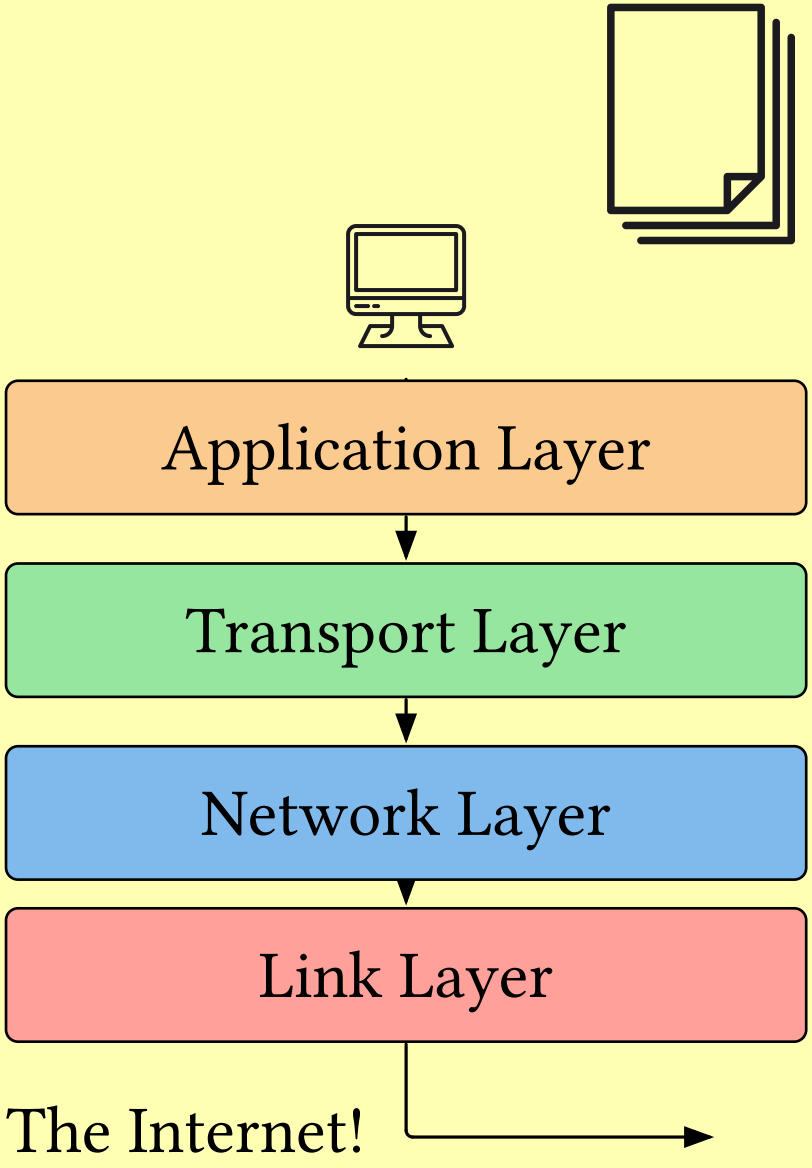


The TCP/IP Stack

At the other end, that whole process occurs in reverse.

Rather than adding data, each layer will 'strip' its relevant data.

The program on the other end doesn't actually know that any transfer has happened, it's like the two programs are talking directly to each other!

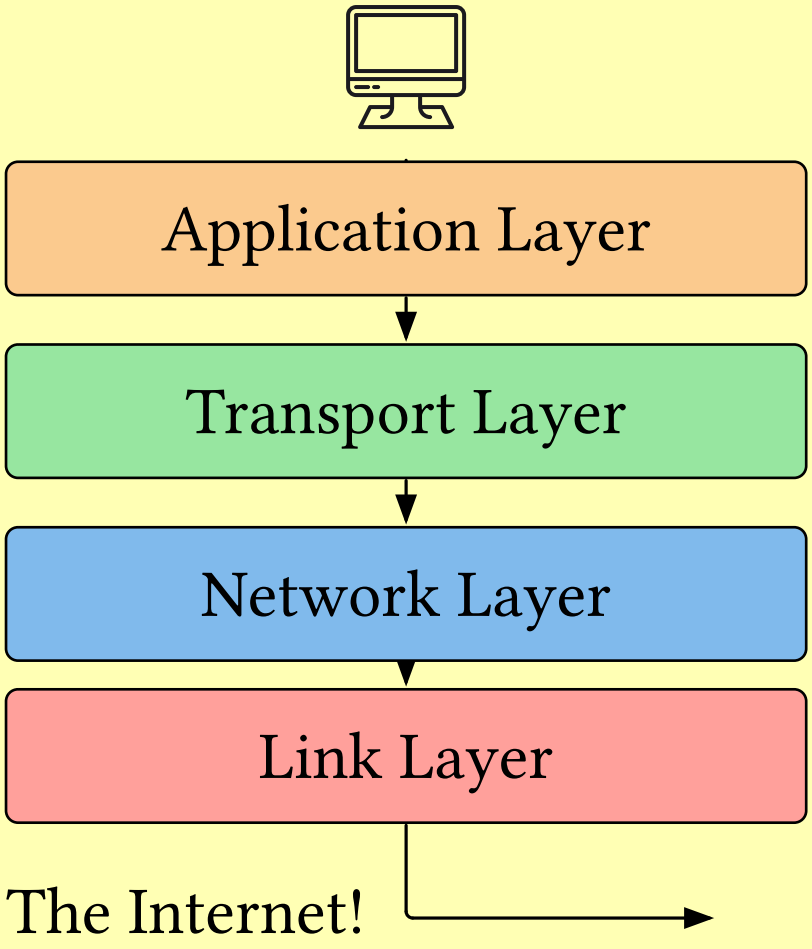


The TCP/IP Stack

What do the four layers do?

Write down what you can remember from the prework.

Then share it with the people near you. Add to your lists!

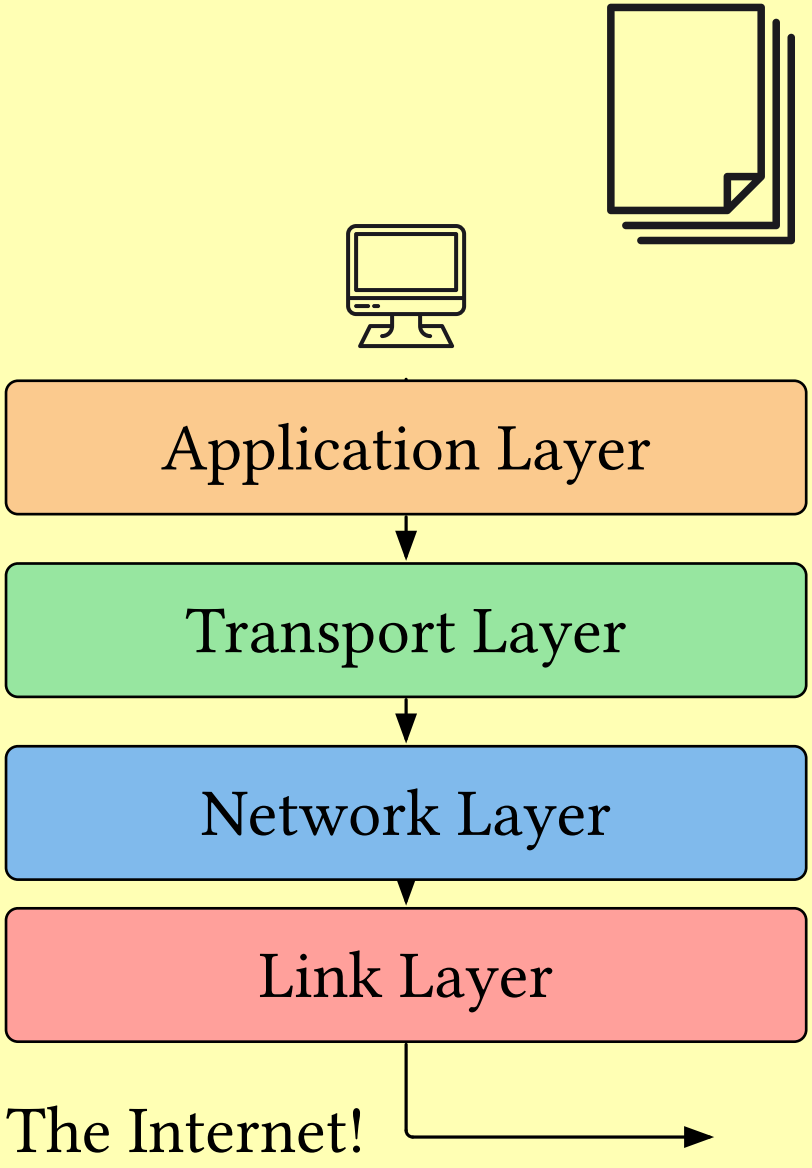


The TCP/IP Stack

Application Layer

Uses protocols designed to work with specific types of applications.

These handle the interaction with the user
(e.g. HTTP, IMAP, FTP)



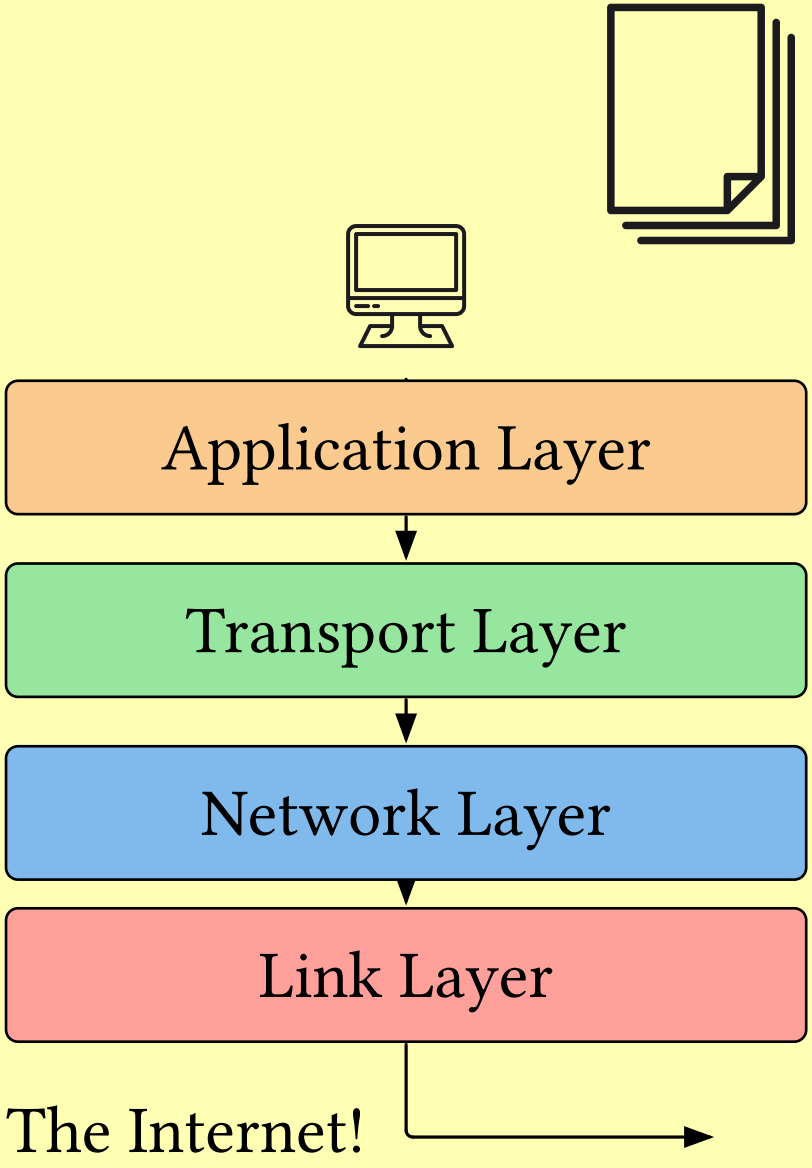
The TCP/IP Stack

Transport Layer

Breaks the data down into segments.

Allocates sequence numbers and source / destination port numbers

(TCP only) Acknowledges delivery of segments

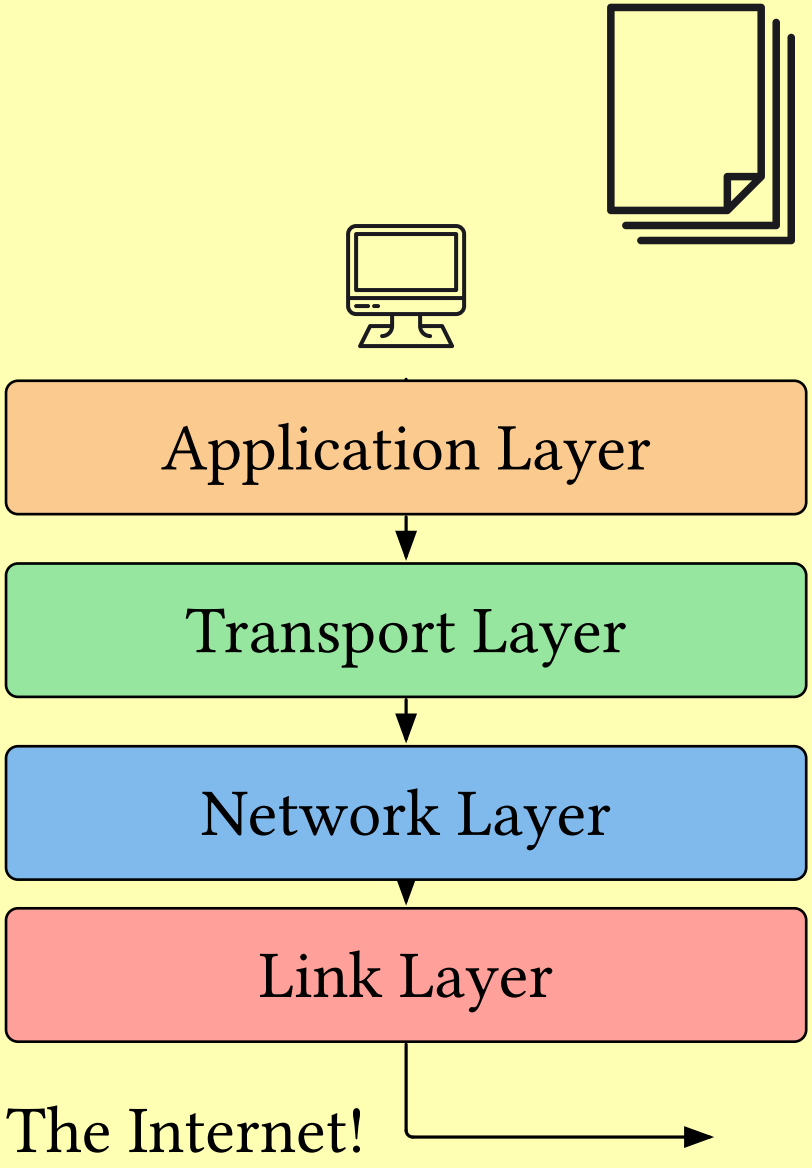


The TCP/IP Stack

Network Layer

Prepares packets for the Internet.

Source and Destination IP Addresses are added

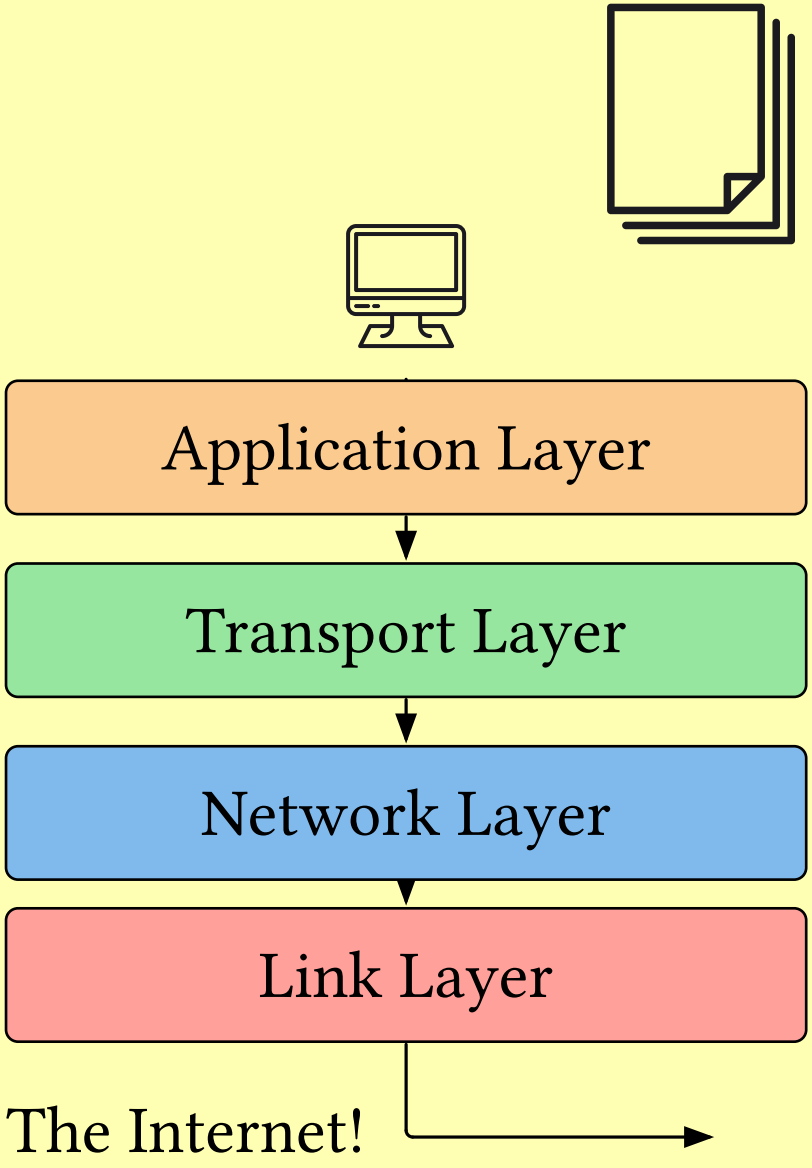


The TCP/IP Stack

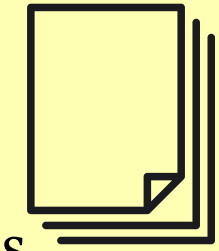
Link Layer

Responsible for transporting the frames across each individual link between hops.

Uses MAC addresses for this communication



Encapsulation



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Program Data

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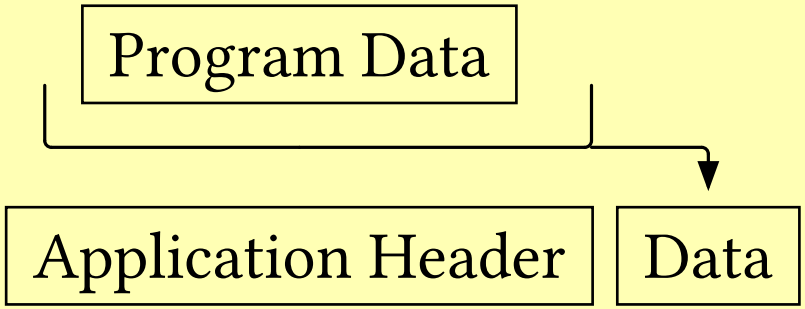

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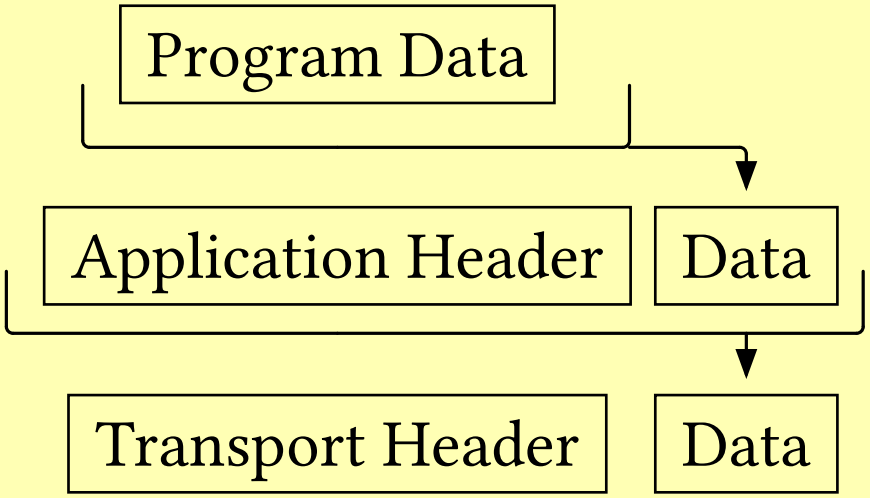

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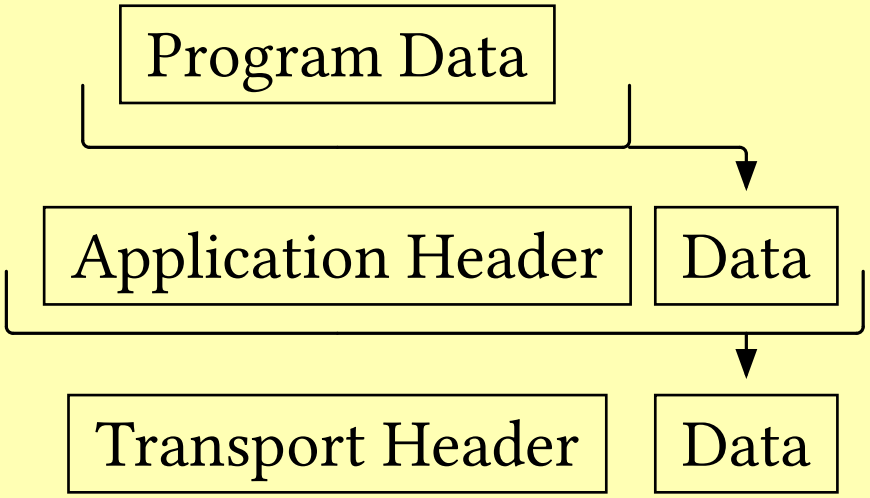
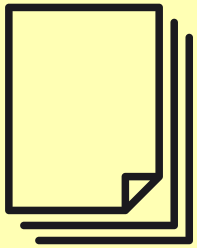


ETC...

Reliable Transport Layer (TCP)

1. The transport layer receives data from the Application layer.

Reliable Transport Layer (TCP)



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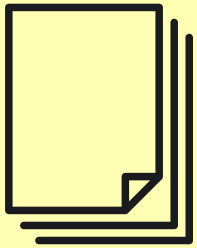

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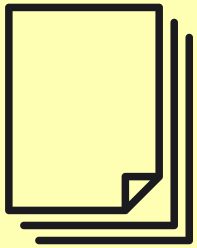

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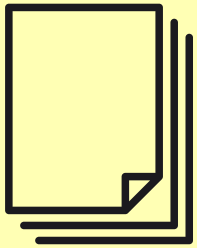

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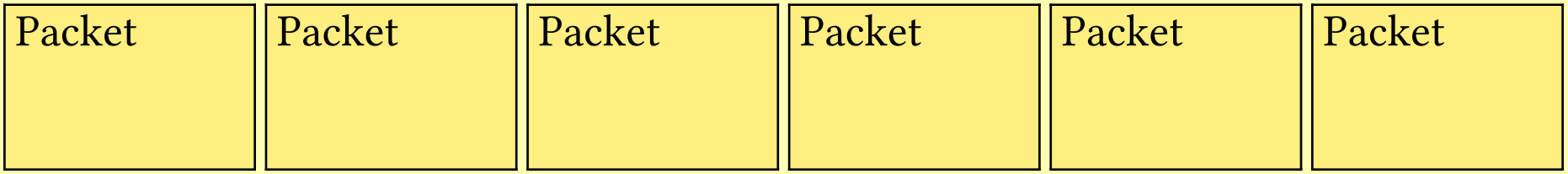

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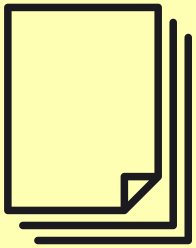

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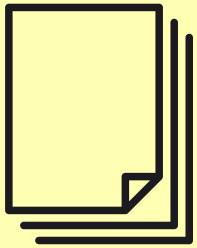

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These packets are then individually sent through the internet! (Packet Switching)

Port Numbers

There are some **well-known port numbers**, which relate to specific applications.

Often, client applications will not use those, and will use what are called **ephemeral port numbers**
(49152 - 65535)

Port Numbers

You need to know a few of the common ones!

Protocol	Use	Port Number
FTP (Control)	File Transfer	20
FTP (Data)	File Transfer	21
SSH	Secure Remote Access	22
SMTP	Mail Transfer	25
HTTP	Website Access	80
POP	Mailbox Access	110
IMAP	Mailbox Access	143
HTTPS	Secure Website Access	443

Reliable Transport Layer (TCP)

Another job that the transport layer performs is the handling of **acknowledgements**.

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When the packet is received by the intended recipient, **their transport layer** sends an **ACK**.

Reliable Transport Layer (TCP)

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When a packet hasn't been acknowledge within a specified time, the **original host transport layer** resends the packet.

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What is a socket?

What is a socket?

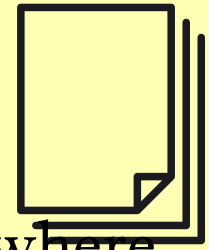
The combination of an **IP Address** and Transport layer **port number**

It represents a **particular process** running on a **particular Internet host** (computer)

e.g.

`38.160.150.31:80`

Network Layer



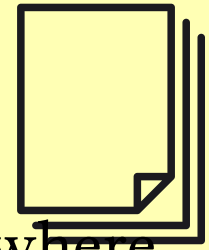
Each Transport Layer packet is passed to the Network Layer, where the Network Header is attached.

This includes the **Source** and **Destination IP Addresses**, as well as a checksum.

1. What is a checksum?

2. How does it know the Destination IP Address?

Network Layer



Each Transport Layer packet is passed to the Network Layer, where the Network Header is attached.

This includes the **Source** and **Destination IP Addresses**, as well as a checksum.

1. What is a checksum?

A value calculated based on the payload, used to confirm that the data hasn't been damaged.

2. How does it know the Destination IP Address?

Link Layer

Each packet is then passed to the Link Layer, where the Link Header is added!

This includes the Source and Destination **MAC Addresses**. These are physical addresses for the host and a device it is directly connected to.

The Link Layer handles each **individual hop** between routers. (Packet Switching!!)

All together now!

Can you remember all of the steps?

L128 - Networks 2 Process now on Google Classroom!